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Wind energy development alters breeding habitat of the Bicknell's Thrush, a threatened range-restricted boreal songbird

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SCHOLARONE MANUSCRIPTS

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Abstract

1) Wind energy development is widely promoted as a key strategy for mitigating climate change. Nonetheless, its local implementation can create ecological trade-offs where energy development overlaps with habitats supporting biodiversity. Understanding how wind energy infrastructure affects breeding habitat for species with restricted distributions is therefore essential for reconciling climate mitigation with biodiversity conservation.
2) We evaluated how the development of four wind farms in a mountainous landscape of eastern Québec (Canada) affects breeding habitat for the globally vulnerable Bicknell's Thrush (*Catharus bicknelli*). Building on the first published high-resolution habitat occupancy model for the Bicknell's Thrush in its breeding range in Québec, we reinterpolated pre-development occurrence probability in areas where infrastructure had been built prior to the model's development. We then quantified impacts on two classes of breeding habitat priority derived from the model (mitigation and protection) using landscape metrics and patch-size distributions. Finally, spatial alternatives under different development scenarios were evaluated by combining wind potential, habitat priority, and habitat aggregation.
3) Wind energy infrastructure reduced the amount of predicted breeding habitat and altered its spatial configuration, with impacts increasing from lower- to higher-priority breeding habitats. Development reduced the

contribution of large habitat patches and reorganized the landscape into smaller and structurally simpler habitat units.

4) Clear spatial trade-offs emerged between wind potential and habitat priority. Impacts varied strongly between wind farms, with the only built project generating the largest share of habitat loss due to its overlap with higher-priority breeding habitat. Under an equally balanced scenario between wind opportunity and ecological stake, spatial analysis indicated that the landscape could theoretically accommodate development in eight times the total area of all four wind farms before substantial impacts on the protection habitat class occur.

5) ***Synthesis and applications.*** Integrating high-resolution species-specific habitat occupancy information as early as possible in the planning process may reduce conflicts between renewable energy development and biodiversity conservation, particularly when development occurs within landscapes supporting species of conservation concern. This work allows stakeholders to better identify wind farm configurations which utilize the best trade-offs between the conservation of sensitive habitats and high-quality locations for renewable energy production.

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Keywords

Bicknell's Thrush, Biodiversity conservation, Habitat loss, High-altitude boreal forest ecosystems, Renewable energy planning, Vulnerable species, Wind farms.

1. Introduction

From resource extraction to land conversion, anthropogenic development is one of the primary drivers of ecosystem transformation worldwide (Steffen *et al.,* 2015), threatening biodiversity (Newbold *et al.,* 2015; Venter *et al.,* 2016) and undermining the functioning of the natural systems on which human societies depend (Costanza *et al.,* 2014; Hernández-Blanco *et al.,* 2022).

One of the clearest expressions of this development is the expansion of infrastructure (Thacker *et al.,* 2019), driven by diverse needs such as energy production (Tong *et al.,* 2019; Kemfert *et al.,* 2022), transportation (van der Ree *et al.,* 2015), urban growth (Güneralp *et al.,* 2020; Li *et al.,* 2022), and access to natural resources (Bridge 2009; Houballah *et al.,* 2021). This expansion frequently results in habitat conversion, degradation, or fragmentation (Li and Yu 2018; van Vliet 2019; Betts *et al.,* 2022), with cascading effects on ecosystem processes and long-term resilience (McWilliams *et al.,* 2019; Grantham *et al.,* 2020). These consequences are well documented: habitat loss and fragmentation can accelerate local population declines (Heinrichs *et al.,* 2016), disrupt gene flow (Stevens *et al.,* 2018), and ultimately constrain species' ability to adjust their distributions in response to climate change (*e.g.*, Holyoak and Heath 2015).

In this context of increasing pressures and intensified competition for space, tensions between anthropogenic development and ecological conservation are becoming increasingly common (Liu *et al.,* 2017; Bowler *et al.,* 2020). Land-use decisions often involve difficult trade-offs between essential ecological components and fundamental human needs (Girardello *et al.,* 2019). However, although long perceived as conflicting objectives, development and conservation can generate joint benefits when integrated through effective spatial planning that reconciles sustainable resource use with ecosystem preservation (Neubert *et al.,* 2025). To date, a substantial body of research has demonstrated that biodiversity – including its taxonomic, functional, and phylogenetic dimensions – enhances ecosystem functioning (*e.g.*, primary production, decomposition, nutrient cycling, trophic interactions) and, consequently, underpins a wide range of ecosystem services such as food production, climate regulation, pest control, and pollination (Cardinale *et al.,* 2012). As such, understanding how anthropogenic pressures affect biodiversity is central not only to species conservation, but also to the maintenance of ecosystem functioning and resilience.

It was long assumed that the loss of rare species would have limited effects on ecosystem functioning due to their low abundance within ecological communities (Grime 1998; Schwartz *et al.,* 2000). However, empirical evidence has since challenged this assumption, showing that the loss of rare species can significantly alter ecosystem processes (Lyons *et al.,* 2005; Bracken and Low 2012). Owing to their distinct and often unique functional traits, rare species frequently perform complementary roles to those of more common species, thereby increasing

functional diversity and contributing key traits that help sustain ecosystem processes under uncertain and changing environmental conditions (Mouillot *et al.,* 2013; Kunze *et al.,* 2025). Moreover, birds play a major role and represent one of the most diverse groups of ecosystem service providers, through functions such as seed dispersal, pollination, nutrient transfer and cycling, trophic regulation, pest control, and ecosystem engineering (Şekercioğlu 2006). Consequently, preventing further declines in bird species with restricted distributions or low population sizes has become a priority conservation objective, as it directly contributes to maintaining functional diversity and enhancing the adaptive capacity of ecosystems in the face of global change (Şekercioğlu *et al.,* 2004; Cooke *et al.,* 2019; Stewart *et al.,* 2022).

The Bicknell's Thrush (*Catharus bicknelli*) is an elusive migratory songbird classified as globally Vulnerable (BirdLife International 2020) and listed as a Threatened species in Canada under the Species at Risk Act (Government of Canada 2012). The species has one of the smallest breeding ranges of North American forest passerines. It reproduces in high-elevation forests across the northeastern part of the continent and its breeding habitat is dominated by balsam fir (*Abies balsamea*) (Townsend *et al.,* 2020). Canada is estimated to support approximately 95% of the species' global breeding range, with nearly 75% located in the province of Québec alone, conferring a substantial share of the global conservation responsibility on this province (COSEWIC 2022). Within its breeding range, the Bicknell's Thrush faces significant threats from habitat loss, primarily due to natural resource extraction and forest management practices, and more

111 recently, wind energy development has emerged as an additional threat to the
112 species (COSEWIC 2022). Although wind energy development is widely
113 recognized to contribute to climate change mitigation by reducing greenhouse gas
114 emissions (Barthelmie and Pryor 2014), its local implementation can create
115 ecological trade-offs through infrastructure expansion (Katzner *et al.,* 2025). In the
116 province of Québec, mitigation strategies often rely on micro-siting, which involves
117 positioning turbines to avoid sensitive habitats (MDDEFP 2013; Lemaître and
118 Lamarre 2020). However, there is a particularly high risk that suitable breeding
119 habitat for the Bicknell's Thrush overlaps with areas of strong wind energy
120 potential, as ridge-top forests are typically more exposed to high wind speeds and
121 thus constitute prime areas targeted for turbine siting. This spatial convergence
122 raises important challenges for reconciling conservation objectives for this
123 vulnerable species (ECCC 2020) with renewable energy development.

124 In this study, we expand on the recently published high-resolution habitat
125 occupancy model by Tremblay *et al.,* (2025), which predicts the probability of
126 occurrence of the Bicknell's Thrush during the breeding period in a mountainous
127 region of eastern Québec. Habitat occupancy models are now essential tools in
128 conservation planning, as they provide spatial predictions of habitat occupancy
129 using environmental variables known to influence species distributions (Guisan
130 and Thuiller 2005; Elith and Leathwick 2009).

131 Building on these predictions, we assess the extent to which ongoing wind energy
132 development affects breeding habitat for the Bicknell's Thrush. We also explore
133 whether integrating species-specific habitat occupancy probability with spatial data

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on wind energy potential can delineate trade-off areas where development pressure on higher-priority habitats may be minimized. We hypothesize that wind energy development disproportionately affects breeding habitat relative to its availability across the landscape. We further expect trade-off areas to emerge where wind potential is high and occupancy probability is comparatively low. Identifying such areas would provide a basis for more informed spatial planning and transparent conciliation between renewable energy development and habitat conservation objectives.

2. Materials and Methods

2.1. Study area

Our study was conducted within the Seigneurie de Beaupré, a privately owned landholding of approximately 1,200 km^2 located in the Massif du Lac Jacques-Cartier, in eastern Québec, Canada (Fig. 1). The area lies northeast of Québec City and spans a mountainous landscape ranging from approximately 100 to over 1,100 m in elevation.

Characterized by extensive forest cover and limited urban development, the Seigneurie de Beaupré forms part of the eastern balsam fir/white birch (*Betula papyrifera*) bioclimatic domain (MFFP, 2021). The region provides suitable high-elevation breeding habitat for the Bicknell's Thrush and lies within the core of the species' range known in the province of Québec.

In recent years, the area has also become a focal point for wind energy development. Since 2013, multiple wind farms have been commissioned within the

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Seigneurie (Fig. 1), making it one of the largest clusters of wind power infrastructure in Canada (Dehghani-Sanij *et al.,* 2022). In addition to the existing North wind farm infrastructure (364 MW), three additional wind farm projects totaling 1,600 MW are currently under review or planned for operation by 2028, bringing the total to four distinct wind farms in the study area (Table S1), hereafter referred to as the North, South, East, and West wind farms. While only the North wind farm is currently operational, the remaining three are expected to become operational in the coming years. Given this expansion and its potential spatial overlap with predicted breeding areas for the Bicknell's Thrush, the region offers a valuable case study for evaluating trade-offs between the conservation of key breeding habitats of species at risk and renewable energy planning.

<Figure 1>

2.2. Habitat occupancy model

To estimate the Bicknell's Thrush breeding habitat priority, we relied on the high-resolution habitat occupancy model developed by Tremblay *et al.* (2025). This model uses logistic regression to estimate the probability of occurrence of the species at a 10 m spatial resolution, based on LiDAR-derived forest structural metrics and topographic variables. Predictor variables include elevation, mean canopy height (adjusted to account for year of LiDAR acquisition), canopy height variability, understory fractional cover as a proxy for stem density, and the proportion of balsam fir.

In their study, two statistical thresholds were derived to guide decision-making: a protection threshold and a mitigation threshold. The protection threshold was defined to maximize specificity (0.95), thereby minimizing false positives and identifying areas where species occurrence is predicted with high confidence. The mitigation threshold was defined to balance sensitivity and specificity (0.80), identifying areas where the predicted probability of occurrence is sufficiently elevated for mitigation measures to be considered, while acknowledging the trade-off between detection and false positive predictions.

Together, these two thresholds allow us to distinguish areas requiring strict protection from those where mitigation-oriented planning may be considered when assessing the spatial distribution of wind energy infrastructure impacts (Fig.1).

More detailed description of model calibration, validation procedures, and threshold selection can be found in Tremblay *et al.* (2025).

2.3. Wind farm geometries

Exact geometries for the wind farm currently in operation (North) were delineated from satellite imagery (Fig. 1). For the remaining wind farms (East, West, and South) that are not operating yet, geometries were reconstructed from publicly available planning maps contained in their respective *Environmental Impact Assessment* (EIA) (BAPE 2024 & 2025, Pesca Environnement 2025). These maps were georeferenced and digitized in ArcGIS Pro 3.6.2 (Fig. 1). Turbine pads were represented as circular features with an area of 1.2 ha (equivalent radius of ~62 m), and access-road corridors were modeled with a total width of 40 m,

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consistent with the already built North wind farm (Table S1). To account for the uncertainty associated with the 10 m resolution habitat occupancy model, all infrastructure features were buffered by an additional 10 m prior to rasterization. Buffered geometries were merged and rasterized to the 10 m grid of the habitat occupancy model to generate the wind farm geometry layer used in subsequent impact analyses.

2.4. Pre- and post-development conditions

Pre-development conditions were defined as an occurrence probability raster free of any wind farm geometries, representing estimated habitat availability prior to wind energy development across the study area. Since the North wind farm was built prior to model development, the original occurrence probability raster reflects post-development conditions within this area. Consequently, reconstruction of pre-development conditions was required (Section 2.4.1). Outside the North wind farm, predicted occurrence probabilities from the original habitat occupancy model (Section 2.2) were retained unchanged, as the remaining wind farms are not yet constructed.

The resulting composite raster represents reconstructed pre-development occurrence probabilities. This continuous raster was subsequently classified into three categories (Fig. S1): non-habitat (below mitigation threshold), mitigation areas (between mitigation and protection thresholds), and protection areas (above the protection threshold). Within the Seigneurie de Beaupré, mitigation areas represented 6,805 ha (4.3% of the Seigneurie), while protection areas represented 1,926 ha (1.2%) (Σ = 8,731 ha; 5.5% of the Seigneurie). Post-development

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conditions were derived by overlaying the combined buffered infrastructure geometries of all wind farms on the classified pre-development occurrence probability raster: pixels landing in the infrastructure geometries were reassigned to the non-habitat class (class 0), while all other pixels retained their original classification (Fig. S1).

2.4.1. Habitat reinterpolation in the North wind farm area

To reconstruct pre-development habitat occupancy probabilities in the North wind farm area, we reinterpolated occurrence probability values inside the existing infrastructure geometries using inverse distance weighting (IDW; R package *gstat* v2.1.4, Pebesma, 2004). Reinterpolation was restricted to pixels located within the buffered geometries and above an elevation threshold of 600 m, corresponding to the lower elevational range typically associated with suitable breeding habitat in the underlying habitat occupancy model (Tremblay *et al.*, 2025). This restriction limited interpolation to ecologically plausible areas where breeding individuals were most likely present prior to development.

To parameterize the IDW interpolation, we optimized the power parameter (β; idp in *gstat*) and the maximum number of neighbors (n_{max}) using leave-one-out cross-validation (LOOCV; Hartmann *et al.*, 2023) on a random subset of 1,000 reference points. A two-step grid search was conducted (Fig. S2), consisting of an initial coarse exploration followed by a refined search around the parameter space minimizing root mean square error (RMSE). The parameter combination minimizing RMSE was retained for final interpolation.

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The optimized IDW model was applied to generate a reconstructed occurrence probability raster in the North wind farm geometries, which was then integrated into the landscape-scale pre-development raster described above.

2.5. Quantification of wind farm development impacts on habitat

The classified pre-development and post-development occurrence probability rasters were compared to quantify the impacts of wind farm development on the predicted Bicknell's Thrush habitat. We computed class-level landscape metrics using the *landscapemetrics* R package v2.2.1 (Hesselbarth *et al.*, 2019), thereby quantifying changes in habitat amount and spatial configuration using three different categories of metrics (fragmentation, connectivity, and heterogeneity) for each habitat class (Table 1). For each metric and habitat class, changes were calculated relative to pre-development conditions and expressed as percentages, to compare across metrics. Patches were delineated using the eight-neighbour (Queen's case) adjacency rule, whereby pixels were considered part of the same patch if they shared either a side or a corner.

<*Table 1*>

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2.6. Habitat patch-size distribution

Landscape metrics related to the habitat class level were complemented by an analysis of habitat patch-size distributions to compare the relative contribution of small and large habitat patches under pre- and post-development conditions. Patch areas (AREA; ha) were extracted using *landscapemetrics* for the two habitat classes of interest (mitigation and protection). For each habitat class and state (pre- and post-), we estimated the distribution of $\log_{10}$-transformed patch size, weighting each patch by its area so that larger patches contributed proportionally to their surface.

We then summarized patch-size distributions using the area-weighted mean patch size: $\mu_w = \Sigma\, a^2 / \Sigma\, a$, where a represents individual patch areas (ha). This metric corresponds to the expected patch size to which a randomly selected hectare of habitat belongs, within the habitat class of interest.

2.7. Evaluating spatial alternatives to minimize habitat impacts

To explore lower-impact development scenarios, we constructed a constraint space in which each pixel of the study area was evaluated according to three dimensions of interest: wind energy potential, habitat priority, and local habitat aggregation (used here as a proxy for habitat functionality, see below), all normalized to a [0,1] interval to facilitate comparability. To restrict the analysis to terrestrial environments, all water bodies were excluded prior to further processing using the official Québec COTQ dataset (MRNF, 2025), available at the same 10 m

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283 spatial resolution as the Bicknell's Thrush habitat occupancy model described in
284 Section 2.2.

285 Wind energy potential was derived from the Global Wind Atlas v4.0 (Davis *et al.*,
286 2023) at 150 m above ground level, corresponding to the approximate hub height
287 of the wind turbines installed in the study area (BAPE, 2024). Pixels with wind
288 speeds below 8 $m \cdot s^{-1}$, the threshold considered technically exploitable (Hélimax,
289 2005), were excluded from further analysis. To represent wind energy potential as
290 a continuous variable rather than a binary constraint, wind speeds ≥ 8 $m \cdot s^{-1}$ were
291 transformed using an affine scaling function. Values were linearly rescaled
292 between the exploitable threshold (8 $m \cdot s^{-1}$) and an upper reference corresponding
293 to the highest 10% of wind speeds in the study area (Davies *et al.*, 2023), above
294 which wind opportunity was set to 1.

295 As described in Sections 2.2. and 2.4., occurrence probability was represented as
296 a continuous raster under pre-development conditions. Thus, to incorporate this
297 information into the constraint space, we transformed occurrence probability using
298 a symmetric logistic function (Godeau *et al.*, 2020). This function was centered on
299 the mitigation threshold which acted as the inflection point, and used a scale
300 parameter of 0.03 to produce a gradual increase in habitat priority with increasing
301 predicted occurrence probability.

302 Finally, to account for habitat functionality, spatial context was incorporated
303 through a measure of local aggregation. Functionality was approximated as the
304 local proportion of pixels exceeding the mitigation threshold within a fixed square
305 moving window of 37 x 37 pixels (~ 13.7 ha at 10 m resolution), aligning with the

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306 mean home-range size of the female Bicknell's Thrush (95% fixed kernel; Aubry
307 *et al.*, 2011).

308 The three normalized dimensions were then integrated into a composite score as
309 a linear combination of wind opportunity, habitat priority, and habitat aggregation.
310 Wind energy potential contributed positively to the score, whereas habitat priority
311 and local aggregation contributed negatively, reflecting increasing ecological cost.
312 The resulting raw score was subsequently rescaled to the [0,1] interval using min-
313 max normalization across the study area. Higher values therefore identify locations
314 where wind opportunity is high and ecological constraints are comparatively low,
315 while lower values indicate areas where development would likely entail stronger
316 ecological costs.

317 Because the choice of weights can influence the ranking of locations in multi-
318 criteria analyses (Adem Esmail and Geneletti, 2018), we explored three weighting
319 schemes representing different planning perspectives (Table 2). In the reference
320 scenario, the three dimensions were combined using weights defined to reflect a
321 neutral planning perspective (50-50), in which wind development potential and
322 ecological constraints were given equal overall importance (because two of the
323 three dimensions represent ecological information, weights were distributed
324 across habitat priority and habitat aggregation such that their combined
325 contribution balanced that of wind opportunity). Additional scenarios included a
326 development-oriented configuration giving greater weight to wind potential, and a
327 conservation-oriented configuration emphasizing ecological constraints. Both
328 configurations are inversely symmetric weighting schemes (70-30 vs. 30-70

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between development and ecological objectives). Comparing the spatial outcomes across these alternative scenarios allowed us to illustrate how the resulting prioritization varies under plausible differences in planning preferences.

<Table 2>

Ultimately, we used the ranking produced by the constraint space to estimate potential alternatives for wind farm placement. The total area of the four wind farms considered in this study (3,691 ha) was used as a reference unit of development. For each scenario, pixels are selected from most suitable to least suitable until the reference area of development is reached. We then iterate by progressively adding one additional reference area at a time. Therefore, the cumulative allocated area A at the ith iteration is $A_i = i \times 3{,}691\ \mathrm{ha}$. This procedure allowed us to estimate how much area could potentially be allocated to wind farm development before affecting habitat meeting the mitigation and protection classes.

This approach was intended to evaluate whether substantial development could be accommodated while maintaining limited impacts on high-priority habitat. We subsequently focused on the reference scenario to illustrate this potential under a balanced planning perspective. A theoretical available development area was defined as the cumulative area at the iteration which began to affect the protection class. To mimic development, we overlaid the theoretical development area on the pre-development probability raster. Similarly to Section 2.4, pixels landing in the theoretical development area were reassigned to the non-habitat class (class 0), while all other pixels retained their original classification.

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Finally, the same landscape metrics analysis used for the original wind-farm geometries (Section 2.5) was applied to quantify the resulting changes in habitat composition and configuration for this theoretical development area.

3. Results

3.1. Habitat reinterpolation in the North wind farm area

Based on LOOCV optimization, the parameter combination β (idp) = 1.85 and n_{max} = 14 produced the lowest cross-validated RMSE (0.0506; Fig. S2). Within the interpolated area, 49.8% of pixels reached or exceeded the mitigation threshold and 17.0% the protection threshold (respectively 329 ha and 170 ha; see Fig. S3 for visual comparison of model predictions before and after IDW reinterpolation).

3.2. Impacts of wind energy development on the Bicknell's Thrush habitat

Landscape metrics revealed that wind farm development reduced total habitat area and degraded habitat configuration, with the protection class consistently more affected than the mitigation class (Fig. 2A). Total mitigation class habitat area declined by 11.1% and its mean patch size decreased by 10.4%. Similarly, total protection class habitat area declined by 17.1% and its mean patch size decreased by 14.4%. Metrics capturing the availability of large, internally intact habitat showed the greatest losses: total core area (TCA) decreased by 24% and 44.3%, largest patch index (LPI) fell by 72% and 79%, and effective mesh size (MESH) by 71.5% and 84.9%, for the mitigation and protection classes respectively. Separating each wind farm's contribution, we found that impacts were predominantly driven by the

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373 North wind farm (57.1% of the total losses across all combined metrics), followed
374 by the West wind farm (34.8%), with comparatively minor contributions from the
375 East (5.1%) and the South (3%) wind farms.

376 <*Figure 2*>
377

378 Area-weighted patch-size distributions showed a clear shift toward smaller patches
379 after development (Fig. 3). Large habitat mitigation class patches accounted for a
380 smaller proportion of the total habitat area, with the distribution shifting toward
381 smaller units (μ_w decreased from 98.1 to 31.5 ha; -68%). This pattern was more
382 pronounced for the protection class, with a greater decline in the contribution of
383 large patches and a greater dominance of small habitat patches (μ_w decreased
384 from 44.2 to 8.1 ha; -82%).

385 <*Figure 3*>
386

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3.3. Identifying spatial configurations with lower impacts

For each scenario, the construction of the constraint space produced a continuous ranking of all wind-eligible pixels according to their relative suitability for development under the combined ecological and wind constraints (Fig. 4). The resulting score ranged from 0 to 1, ordering all pixels from the least to the most suitable locations for wind farm development.

<Figure 4>

As expected, when iterating through the allocation of the original total wind farm area across the ranked landscape, cumulative habitat loss increased in all scenarios (Fig. 5). However, the development scenario consistently produced the highest cumulative habitat loss at a given iteration. Habitat loss appeared earliest in this scenario: at iteration 2 for the mitigation class (232 ha) and at iteration 3 for the protection class (28 ha). Conversely, the conservation scenario consistently delayed the onset of habitat loss relative to the other scenarios: losses remained null until iteration 8 (1 ha) for the mitigation class, and no habitat loss occurred until iteration 15 (434 ha) for the protection class.

<Figure 5>

Finally, the reference scenario was an intermediate between the other two scenarios: habitat loss started at iteration 3 (27 ha) for the mitigation class and losses remained null until iteration 8 (8 ha) for the protection class. This 8th iteration corresponded to the theoretical available development area defined in Section 2.7.

410 for subsequent analysis. In other words, this theoretical development area is eight
411 times larger than the original total wind farm area. At this level of development,
412 landscape metrics indicated relatively limited structural changes, with impacts
413 driven primarily by changes in the mitigation habitat class (Fig. 2B). The direction
414 and magnitude of change in landscape metrics between the original total wind farm
415 area and this theoretical development area are illustrated in Fig. 2C.

4. Discussion

4.1. Effects of wind energy development on Bicknell's Thrush habitat

Understanding the magnitude and patterns of habitat alteration remains a critical step toward evaluating the implications of wind energy development for a species at risk (*e.g.*, Kerr and Deguise, 2004; Fischer and Lindenmayer, 2007), which is of high priority for the Bicknell's Thrush given its status and its strong dependence on structurally complex high-elevation forest habitat (Townsend *et al.*, 2020). As such, the results support both hypotheses of this study: 1) wind energy development disproportionately affects breeding habitat for the Bicknell's Thrush relative to its availability across the study area, and 2) clear spatial trade-offs emerge between wind potential and occupancy probability.

The impact assessment indicates that development affects several dimensions of habitat amount and configuration. The dominant process induced by wind farm development is not the widespread creation of new habitat fragments but rather a reduction in predicted habitat patch size. This pattern is reflected by a strong decline in the largest predicted habitat patches (LPI) and the associated reduction in area-weighted mean patch size, indicating that large habitat units contribute substantially less to the landscape after development. In contrast, the number of patches (NP) does not increase substantially, further supporting the interpretation that development primarily reduces the size of existing habitat patches rather than generating numerous new fragments. This shift is particularly pronounced for the protection habitat class. In parallel, total predicted habitat area (CA) for the

Bicknell's Thrush also declines, and several indicators point to a simplification of predicted habitat configuration. The variability in patch area (AREA_CV) and shape (SHAPE_CV) decreases following development, suggesting a loss of large habitat units and a tendency toward smaller and more homogeneous patches. These effects are evident for both mitigation and protection habitat classes but remain consistently more pronounced for the protection class, which represents the most important habitat for the preservation of the species. Taken together, these changes are consistent with a growing body of evidence indicating that habitat alteration induced by wind energy infrastructure can alter its ecological functioning, extending across taxa and ecological processes (Kuvlesky Jr. *et al.*, 2007; Northrup and Wittemyer, 2013). Previous studies have, for example, documented functional habitat loss for migratory soaring birds (Marques *et al.*, 2020), conflicts between wind energy production in forests and tree-roosting bats (Reusch *et al.*, 2023), and cascading ecological impacts across trophic levels in ecosystems surrounding wind farms (Thaker *et al.*, 2018).

From a population perspective, the reduction in predicted habitat patch size observed in the present study may nonetheless indicate increasing vulnerability of local breeding populations, as landscape contexts with smaller patches are associated with higher local extinction probabilities in Bicknell's Thrush populations (McFarland *et al.*, 2008). Here, because the largest habitat patches can decrease in size while remaining relatively large, demographic consequences may remain more moderate for habitats meeting the mitigation threshold. In contrast, habitats meeting the protection threshold tend to occur in smaller

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462 patches, where further reductions in patch size may be more likely to coincide with
463 higher extinction risk. In the present study, mitigation and protection habitats were
464 analyzed as separate classes. As a result, the structural changes documented
465 here do not account for habitat patches composed of a mixture of both habitat
466 classes, meaning that our analysis may not fully capture the structural conditions
467 experienced by breeding individuals. In addition, the McFarland *et al.* (2008)
468 relationship originates from an unpublished dataset collected in a different region
469 and should therefore be considered indicative rather than predictive. Re-examining
470 the relationship between habitat patch size and local extinction probability across
471 the breeding range of the Bicknell's Thrush would therefore represent an important
472 avenue for future research.

473 More broadly, as reviewed by Venier *et al.* (2014), our results align with evidence
474 that natural resource development in boreal forests primarily alters biodiversity
475 through changes in landscape structure and habitat configuration, including shifts
476 in forest age distribution, stand composition, and patch properties. Rather than
477 causing immediate species loss, these habitat modifications more often alter
478 community composition and population dynamics by reducing the availability and
479 connectivity of suitable habitat patches.

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4.2. Implications for wind energy planning

Between the four wind farms considered, the North wind farm (the only project in operation so far) accounts for the largest estimated impacts on habitat among all projects considered. Notably, this wind farm overlaps with areas supporting extensive predicted breeding habitat and therefore generates a disproportionate share of the total impact, although predicted habitat of the protection class represents only a limited fraction of the study area (1.2%). In this regard, it is noteworthy that the North wind farm was built before the Bicknell's Thrush species distribution model from Tremblay *et al.* (2025) became available, highlighting the value of integrating high-resolution habitat occupancy information as early as possible in the planning process, as advocated by Zurrell *et al.* (2022). At the time of the project assessment conducted prior to its construction, however, no regulatory requirements mandated the inclusion of ecological information on the habitat of the Bicknell's Thrush. In contrast, the more recent wind farm projects in the study area (East, West and South) were assessed under a new regulatory context requiring consideration of species-specific ecological information, already reducing impacts on the species through improved project siting as reflected in their comparatively smaller share of the total predicted impact.

Beyond the assessment of existing developments, our study provides insights for planning wind energy infrastructure while mitigating effects on sensitive breeding habitat. Our demonstration that the study area could accommodate a development capacity with an equivalent of eight times the total area of the four wind farms considered in our study, under a balanced 50-50 scenario, before substantial

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504 impacts to the protection habitat class occur further supports this overall
505 interpretation.

506 Even under a development-oriented scenario giving 70% weight to wind
507 development, losses of protection class habitat remained limited at early iterations,
508 with only 28 ha affected at the third iteration, corresponding to 11 073 ha of wind
509 farm infrastructures. Moreover, since the earliest iterations of the original total wind
510 farm area considered in this study correspond to pixels where wind energy
511 potential is high while ecological constraints remain minimal, the analysis also
512 indicates that many additional pixels remain compatible for development while
513 maintaining limited impacts on habitat of our species at risk. Thus, the landscape
514 offers a wide range of viable development options. This may prove particularly
515 valuable as additional constraints are incorporated into the planning process (see
516 below). It should be noted that, in practice, wind farms are not planned on a pixel-
517 by-pixel basis, as turbine pads, access roads, and associated infrastructure require
518 extensive connected surfaces. Here, development was simulated using an
519 equivalent surface approach, focusing on the total area required by the wind farms
520 considered, rather than constraints on their spatial configuration. This
521 simplification still allowed us to evaluate the trade-offs between wind farm
522 development and habitat conservation of the Bicknell's Thrush.

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4.3. Broader planning constraints and study limitations

Additionally, it is important to recognize that wind farm siting occurs within a multi-constraint decision framework (Wimhurst *et al.*, 2023). In practice, siting decisions unfold within a broader socio-ecological planning context where, beyond technical feasibility and regulatory requirements, development must also account for increasing expectations regarding social acceptability, human settlements, scenic viewpoints, and multiple conservation priorities present within the landscape, for instance the woodland caribou (*Rangifer tarandus caribou*) habitat in our study area (Mackey *et al.*, 2024). Evidence from past Québec wind-energy projects also shows that siting decisions must navigate a complex planning context involving multiple stakeholders and competing land-use priorities (Simard, 2018). These considerations highlight that land-use planning for renewable energy increasingly requires integrating decision-making within a broader social-ecological framework (Fischer *et al.*, 2021; Giakoumi *et al.*, 2025).

We acknowledge that our study focuses on a single ecological stake, *i.e.*, the Bicknell's Thrush breeding habitat. Hence our analysis does not represent the full set of constraints governing wind energy development. However, our framework does offer the flexibility to add other constraints within the n-dimensional constraint space. Additional parameters could be incorporated, either as new axes or within existing dimensions, to better reflect the diversity of factors considered in real-world planning processes. We should keep in mind that, in practice, for a given set of constraints, the geographic space may not necessarily offer satisfying solutions for all parties involved.

547 Finally, it is important to recognize that habitat alteration represents only one of
548 several pathways through which wind energy development can affect biodiversity
549 (Thaker *et al.*, 2018). As reviewed by Katzner *et al.* (2025), wind farm facilities can
550 also generate direct mortality through turbine collisions, induce behavioural
551 responses such as avoidance or attraction, and modify animal movements through
552 barrier effects or displacement. In addition, turbines can alter local environmental
553 conditions, including noise, turbulence and microclimate, which may influence
554 wildlife behaviour and activity patterns. Together, these mechanisms highlight the
555 need for future research to better capture the overall ecological effects of wind
556 energy development on the Vulnerable Bicknell's Thrush.

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References

Adem Esmail, B., & Geneletti, D. (2018). Multi-criteria decision analysis for nature conservation: A review of 20 years of applications. *Methods in Ecology and Evolution*, *9*(1), 42-53. https://doi.org/10.1111/2041-210X.12899

Aubry, Y., Desrochers, A., & Seutin, G. (2011). Response of Bicknell's Thrush (*Catharus bicknelli*) to boreal silviculture and forest stand edges: A radio-tracking study. *Canadian Journal of Zoology*, *89*(6), 474482. https://doi.org/10.1139/z11-011

Barthelmie, R. J., & Pryor, S. C. (2014). Potential contribution of wind energy to climate change mitigation. *Nature Climate Change*, *4*(8), 684-688. https://doi.org/10.1038/nclimate2269

Betts, M. G., Yang, Z., Hadley, A. S., Smith, A. C., Rousseau, J. S., Northrup, J. M., Nocera, J. J., Gorelick, N., & Gerber, B. D. (2022). Forest degradation drives widespread avian habitat and population declines. *Nature Ecology & Evolution*, *6*(6), 709-719. https://doi.org/10.1038/s41559-022-01737-8

Bureau d'audiences publiques sur l'environnement (BAPE). (2024). Rapport 375—Projet éolien Des Neiges – Secteur sud dans la MRC de La Côte-de-Beaupré. https://voute.bape.gouv.qc.ca/dl?id=00000647611

Bureau d'audiences publiques sur l'environnement (BAPE). (2025). Rapport 385—Projet éolien Des Neiges – Secteur Charlevoix à Baie Saint Paul et à Saint Urbain. https://voute.bape.gouv.qc.ca/dl?id=00000723052

COSEWIC. Bicknell's Thrush (*Catharus bicknelli*): COSEWIC assessment and status report 2022. (2023, octobre 18). https://www.canada.ca/en/environment-climate-change/services/species-risk-public-registry/cosewic-assessments-status-reports/bicknell-thrush-2022.html

BirdLife International. (2020). Bicknell's Thrush *Catharus Bicknelli* Species Factsheet. (2025, octobre 8). https://datazone.birdlife.org/species/factsheet/bicknells-thrush-catharus-bicknelli

585 Bowler, D. E., Bjorkman, A. D., Dornelas, M., Myers-Smith, I. H., Navarro, L. M.,
586 Niamir, A., Supp, S. R., Waldock, C., Winter, M., Vellend, M., Blowes, S. A.,
587 Böhning-Gaese, K., Bruelheide, H., Elahi, R., Antão, L. H., Hines, J., Isbell, F.,
588 Jones, H. P., Magurran, A. E., … Bates, A. E. (2020). Mapping human pressures
589 on biodiversity across the planet uncovers anthropogenic threat complexes.
590 *People and Nature*, *2*(2), 380-394. https://doi.org/10.1002/pan3.10071

591 Bracken, M. E. S., & Low, N. H. N. (2012). Realistic losses of rare species
592 disproportionately impact higher trophic levels. *Ecology Letters*, *15*(5), 461-467.
593 https://doi.org/10.1111/j.1461-0248.2012.01758.x

594 Bridge, G. (2009). Material Worlds: Natural Resources, Resource Geography
595 and the Material Economy. *Geography Compass*, *3*(3), 1217-1244.
596 https://doi.org/10.1111/j.1749-8198.2009.00233.x

597 Cardinale, B. J., Duffy, J. E., Gonzalez, A., Hooper, D. U., Perrings, C., Venail,
598 P., Narwani, A., Mace, G. M., Tilman, D., Wardle, D. A., Kinzig, A. P., Daily, G.
599 C., Loreau, M., Grace, J. B., Larigauderie, A., Srivastava, D. S., & Naeem, S.
600 (2012). Biodiversity loss and its impact on humanity. *Nature*, *486*(7401), 59-67.
601 https://doi.org/10.1038/nature11148

602 Ministère des Forêts, de la Faune et des Parcs (MFFP). (2021). Classification
603 écologique du territoire québécois (2e édition). *Direction des inventaires*
604 *forestiers, gouvernement du Québec.*
605 https://mffp.gouv.qc.ca/documents/forets/inventaire/classification_ecologique_ter
606 ritoire_quebecois.pdfCanada, E. and C. C. (2014, août 14). *Species at Risk Act :*
607 *Order amending Schedule 1 (volume 146, number 14, July 4, 2012)*
608 [Assessments;research]. https://www.canada.ca/en/environment-climate-
609 change/services/species-risk-public-registry/orders/schedule-1-volume-146-14-
610 july-2012.html

611 Cooke, R. S. C., Eigenbrod, F., & Bates, A. E. (2019). Projected losses of global
612 mammal and bird ecological strategies. *Nature Communications*, *10*(1), 2279.
613 https://doi.org/10.1038/s41467-019-10284-z

614 Costanza, R., de Groot, R., Sutton, P., van der Ploeg, S., Anderson, S. J.,
615 Kubiszewski, I., Farber, S., & Turner, R. K. (2014). Changes in the global value
616 of ecosystem services. *Global Environmental Change*, *26*, 152-158.
617 https://doi.org/10.1016/j.gloenvcha.2014.04.002

618 Davis, N. N., Badger, J., Hahmann, A. N., Hansen, B. O., Mortensen, N. G.,
619 Kelly, M., Larsén, X. G., Olsen, B. T., Floors, R., Lizcano, G., Casso, P., Lacave,
620 O., Bosch, A., Bauwens, I., Knight, O. J., Loon, A. P. van, Fox, R., Parvanyan, T.,
621 Hansen, S. B. K., … Drummond, R. (2023). The Global Wind Atlas: A High-
622 Resolution Dataset of Climatologies and Associated Web-Based Application.
623 *Bulletin of the American Meteorological Society*, *104*(8), E1507-E1525.
624 https://doi.org/10.1175/BAMS-D-21-0075.1

625 Dehghani-Sanij, A. R., Al-Haq, A., Bastian, J., Luehr, G., Nathwani, J.,
626 Dusseault, M. B., & Leonenko, Y. (2022). Assessment of current developments
627 and future prospects of wind energy in Canada. *Sustainable Energy*
628 *Technologies and Assessments*, *50*, 101819.
629 https://doi.org/10.1016/j.seta.2021.101819

630 Elith, J., & Leathwick, J. R. (2009). Species Distribution Models: Ecological
631 Explanation and Prediction Across Space and Time. *Annual Review of Ecology,*
632 *Evolution, and Systematics*, *40*(Volume 40, 2009), 677-697.
633 https://doi.org/10.1146/annurev.ecolsys.110308.120159

634 Environnement et Changement Climatique Canada (ECCC). (2020). Grive de
635 Bicknell (*Catharus bicknelli*) : Programme de rétablissement 2020.
636 https://www.canada.ca/fr/environnement-changement-
637 climatique/services/registre-public-especes-peril/programmes-
638 retablissement/grive-bicknell-2020.html

639 Fischer, J., & Lindenmayer, D. B. (2007). Landscape modification and habitat
640 fragmentation: A synthesis. *Global Ecology and Biogeography*, *16*(3), 265-280.
641 https://doi.org/10.1111/j.1466-8238.2007.00287.x

642 Fischer, J., Riechers, M., Loos, J., Martin-Lopez, B., & Temperton, V. M. (2021).
643 Making the UN Decade on Ecosystem Restoration a Social-Ecological
644 Endeavour. *Trends in Ecology & Evolution*, *36*(1), 20-28.
645 https://doi.org/10.1016/j.tree.2020.08.018

646 Giakoumi, S., Richardson, A. J., Doxa, A., Moro, S., Andrello, M., Hanson, J. O.,
647 Hermoso, V., Mazor, T., McGowan, J., Kujala, H., Law, E., Álvarez-Romero, J.
648 G., Magris, R. A., Gissi, E., Arafeh-Dalmau, N., Metaxas, A., Virtanen, E. A., Ban,
649 N. C., Runya, R. M., … Katsanevakis, S. (2025). Advances in systematic
650 conservation planning to meet global biodiversity goals. *Trends in Ecology &*
651 *Evolution*, *40*(4), 395-410. https://doi.org/10.1016/j.tree.2024.12.002

652 Girardello, M., Santangeli, A., Mori, E., Chapman, A., Fattorini, S., Naidoo, R.,
653 Bertolino, S., & Svenning, J.-C. (2019). Global synergies and trade-offs between
654 multiple dimensions of biodiversity and ecosystem services. *Scientific Reports*,
655 *9*(1), 5636. https://doi.org/10.1038/s41598-019-41342-7

656 Godeau, U., Bouget, C., Piffady, J., Pozzi, T., & Gosselin, F. (2020). Lack of
657 definition of mathematical terms in ecology: The case of the sigmoid class of
658 functions in macro-ecology. *Ecology and Evolution*, *10*(24), 14209-14220.
659 https://doi.org/10.1002/ece3.7016

660 Grantham, H. S., Duncan, A., Evans, T. D., Jones, K. R., Beyer, H. L., Schuster,
661 R., Walston, J., Ray, J. C., Robinson, J. G., Callow, M., Clements, T., Costa, H.
662 M., DeGemmis, A., Elsen, P. R., Ervin, J., Franco, P., Goldman, E., Goetz, S.,
663 Hansen, A., … Watson, J. E. M. (2020). Anthropogenic modification of forests
664 means only 40% of remaining forests have high ecosystem integrity. *Nature*
665 *Communications*, *11*(1), 5978. https://doi.org/10.1038/s41467-020-19493-3

666 Grime, J. P. (1998). Benefits of plant diversity to ecosystems: Immediate, filter
667 and founder effects. *Journal of Ecology*, *86*(6), 902-910.
668 https://doi.org/10.1046/j.1365-2745.1998.00306.x

669 Guisan, A., & Thuiller, W. (2005). Predicting species distribution: Offering more
670 than simple habitat models. *Ecology Letters*, *8*(9), 993-1009.
671 https://doi.org/10.1111/j.1461-0248.2005.00792.x

672 Güneralp, B., Reba, M., Hales, B. U., Wentz, E. A., & Seto, K. C. (2020). Trends
673 in urban land expansion, density, and land transitions from 1970 to 2010: A
674 global synthesis. *Environmental Research Letters*, *15*(4).
675 https://doi.org/10.1088/1748-9326/ab6669

676 Hartmann, K., Krois, J., Rudolph, A. (2023). Statistics and Geodata Analysis
677 using R (SOGA-R). *Department of Earth Sciences, Freie Universitaet Berlin.*
678 https://www.geo.fu-berlin.de/en/v/soga-r/Advanced-
679 statistics/Geostatistics/Inverse-Distance-Weighting-IDW/index.html

680 Heinrichs, J. A., Bender, D. J., & Schumaker, N. H. (2016). Habitat degradation
681 and loss as key drivers of regional population extinction. *Ecological Modelling*,
682 *335*, 64-73. https://doi.org/10.1016/j.ecolmodel.2016.05.009

683 Hélimax Énergie inc. & AWS Truewind, LLC. (2005, juin). *Inventaire du potentiel*
684 *éolien exploitable du Québec* [Technical Report]. Ministère des Ressources
685 naturelles et de la Faune du Québec (MRNF).
686 https://www.economie.gouv.qc.ca/fileadmin/contenu/documents_soutien/secteur
687 _activites/energie/vent_inventaire_inventaire_2005.pdf

688 Hernández-Blanco, M., Costanza, R., Chen, H., deGroot, D., Jarvis, D.,
689 Kubiszewski, I., Montoya, J., Sangha, K., Stoeckl, N., Turner, K., & van ‘t Hoff, V.
690 (2022). Ecosystem health, ecosystem services, and the well-being of humans
691 and the rest of nature. *Global Change Biology*, *28*(17), 5027-5040.
692 https://doi.org/10.1111/gcb.16281

693 Hesselbarth, M. H. K., Sciaini, M., With, K. A., Wiegand, K., & Nowosad, J.
694 (2019). Landscapemetrics: An open-source R tool to calculate landscape
695 metrics. *Ecography*, *42*(10), 1648-1657. https://doi.org/10.1111/ecog.04617

696 Holyoak, M., & Heath, S. K. (2016). The integration of climate change, spatial
697 dynamics, and habitat fragmentation: A conceptual overview. *Integrative*
698 *Zoology*, *11*(1), 40-59. https://doi.org/10.1111/1749-4877.12167

699 Houballah, M., Mathias, J.-D., & Cordonnier, T. (2021). An Infrastructure
700 Perspective for Enhancing Multi-functionality of Forests: A Conceptual Modeling
701 Approach. *Earth's Future*, *9*(1), e2019EF001369.
702 https://doi.org/10.1029/2019EF001369

703 Katzner, T. E., Nelson, D. M., Marques, A. T., Voigt, C. C., Lambertucci, S. A.,
704 Rebolo, N., Bernard, E., Diehl, R., & Murgatroyd, M. (2025). Impacts of onshore
705 wind energy production on biodiversity. *Nature Reviews Biodiversity*, *1*(9),
706 567-580. https://doi.org/10.1038/s44358-025-00078-1

707 Kemfert, C., Präger, F., Braunger, I., Hoffart, F. M., & Brauers, H. (2022). The
708 expansion of natural gas infrastructure puts energy transitions at risk. *Nature*
709 *Energy*, *7*(7), 582-587. https://doi.org/10.1038/s41560-022-01060-3

710 Kerr, J. T., & Deguise, I. (2004). Habitat loss and the limits to endangered
711 species recovery. *Ecology Letters*, *7*(12), 1163-1169.
712 https://doi.org/10.1111/j.1461-0248.2004.00676.x

713 Kunze, C., Bahlburg, D., Urrutia-Cordero, P., Striebel, M., Kelpsiene, E.,
714 Langenheder, S., Donohue, I., & Hillebrand, H. (2025). Partitioning species
715 contributions to ecological stability in disturbed communities. *Ecological*
716 *Monographs*, *95*(1), e1636. https://doi.org/10.1002/ecm.1636

717 Kuvlesky Jr., W. P., Brennan, L. A., Morrison, M. L., Boydston, K. K., Ballard, B.
718 M., & Bryant, F. C. (2007). Wind Energy Development and Wildlife Conservation :
719 Challenges and Opportunities. *The Journal of Wildlife Management*, *71*(8),
720 2487-2498. https://doi.org/10.2193/2007-248Lemaître, J., & Lamarre, V. (2020).
721 Effects of wind energy production on a threatened species, the Bicknell's Thrush
722 *Catharus bicknelli*, with and without mitigation. *Bird Conservation International*,
723 *30*(2), 194-209. https://doi.org/10.1017/S095927092000012X

Li, G., Fang, C., Li, Y., Wang, Z., Sun, S., He, S., Qi, W., Bao, C., Ma, H., Fan, Y., Feng, Y., & Liu, X. (2022). Global impacts of future urban expansion on terrestrial vertebrate diversity. *Nature Communications*, *13*(1), 1628. https://doi.org/10.1038/s41467-022-29324-2

Lin, Q., & Yu, S. (2018). Losses of natural coastal wetlands by land conversion and ecological degradation in the urbanizing Chinese coast. *Scientific Reports*, *8*(1), 15046. https://doi.org/10.1038/s41598-018-33406-x

Liu, P., Wen, H., Harich, F. K., He, C., Wang, L., Guo, X., Zhao, J., Luo, A., Yang, H., Sun, X., Yu, Y., Zheng, S., Guo, J., Li, L., & Zhang, L. (2017). Conflict between conservation and development: Cash forest encroachment in Asian elephant distributions. *Scientific Reports*, *7*(1), 6404. https://doi.org/10.1038/s41598-017-06751-6

Lyons, K. G., Brigham, C. A., Traut, B. h., & Schwartz, M. W. (2005). Rare Species and Ecosystem Functioning. *Conservation Biology*, *19*(4), 1019-1024. https://doi.org/10.1111/j.1523-1739.2005.00106.x

Mackey, B., Campbell, C., Norman, P., Hugh, S., DellaSala, D. A., Malcolm, J. R., Desrochers, M., & Drapeau, P. (2024). Assessing the Cumulative Impacts of Forest Management on Forest Age Structure Development and Woodland Caribou Habitat in Boreal Landscapes: A Case Study from Two Canadian Provinces. *Land*, *13*(1), 6. https://doi.org/10.3390/land13010006

Marques, A. T., Santos, C. D., Hanssen, F., Muñoz, A.-R., Onrubia, A., Wikelski, M., Moreira, F., Palmeirim, J. M., & Silva, J. P. (2020). Wind turbines cause functional habitat loss for migratory soaring birds. *Journal of Animal Ecology*, *89*(1), 93-103. https://doi.org/10.1111/1365-2656.12961

McFarland, K., Rimmer, C., Frey, S., Faccio, S., & Collins, B. (2008). *Demography, ecology and conservation of Bicknell's Thrush in Vermont, with a special focus on the Northeastern Highlands*. Vermont Center for Ecostudies, Norwich, VT. Technical Report 08-03.

McWilliams, C., Lurgi, M., Montoya, J. M., Sauve, A., & Montoya, D. (2019). The stability of multitrophic communities under habitat loss. *Nature Communications*, *10*(1), 2322. https://doi.org/10.1038/s41467-019-10370-2

Ministère des Ressources Naturelles et des Forêts. Cartographie de l'occupation des terres du Québec, [Jeu de données], dans Données Québec, 2025, mis à jour le 25 mars 2026. https://www.donneesquebec.ca/recherche/dataset/cartographie-de-l-occupation-des-terres-du-quebec.

Mouillot, D., Bellwood, D. R., Baraloto, C., Chave, J., Galzin, R., Harmelin-Vivien, M., Kulbicki, M., Lavergne, S., Lavorel, S., Mouquet, N., Paine, C. E. T., Renaud, J., & Thuiller, W. (2013). Rare Species Support Vulnerable Functions in High-Diversity Ecosystems. *PLOS Biology*, *11*(5), e1001569. https://doi.org/10.1371/journal.pbio.1001569

Newbold, T., Hudson, L. N., Hill, S. L. L., Contu, S., Lysenko, I., Senior, R. A., Börger, L., Bennett, D. J., Choimes, A., Collen, B., Day, J., De Palma, A., Díaz, S., Echeverria-Londoño, S., Edgar, M. J., Feldman, A., Garon, M., Harrison, M. L. K., Alhusseini, T., … Purvis, A. (2015). Global effects of land use on local terrestrial biodiversity. *Nature*, *520*(7545), 45-50. https://doi.org/10.1038/nature14324

Northrup, J. M., & Wittemyer, G. (2013). Characterising the impacts of emerging energy development on wildlife, with an eye towards mitigation. *Ecology Letters*, *16*(1), 112-125. https://doi.org/10.1111/ele.12009

Pebesma, E. J. (2004). Multivariable geostatistics in S: The gstat package. *Computers & Geosciences*, *30*(7), 683-691. https://doi.org/10.1016/j.cageo.2004.03.012

Pesca Environnement. (2025, mars). *Étude d'impact sur l'environnement Volume 2 – Documents cartographiques Partie 1*. https://www.ree.environnement.gouv.qc.ca/dossiers/3211-12-244/3211-12-244-8.pdf

781 Reusch, C., Paul, A. A., Fritze, M., Kramer-Schadt, S., & Voigt, C. C. (2023).
782 Wind energy production in forests conflicts with tree-roosting bats. *Current*
783 *Biology*, *33*(4), 737-743.e3. https://doi.org/10.1016/j.cub.2022.12.050

784 Schwartz, M. W., Brigham, C. A., Hoeksema, J. D., Lyons, K. G., Mills, M. H., &
785 van Mantgem, P. J. (2000). Linking biodiversity to ecosystem function:
786 Implications for conservation ecology. *Oecologia*, *122*(3), 297-305.
787 https://doi.org/10.1007/s004420050035

788 Şekercioğlu, C. H. (2006). Increasing awareness of avian ecological function.
789 *Trends in Ecology & Evolution*, *21*(8), 464-471.
790 https://doi.org/10.1016/j.tree.2006.05.007

791 Şekercioğlu, Ç. H., Daily, G. C., & Ehrlich, P. R. (2004). Ecosystem
792 consequences of bird declines. *Proceedings of the National Academy of*
793 *Sciences*, *101*(52), 18042-18047. https://doi.org/10.1073/pnas.0408049101

794 Simard, L. (2018). Socially Not Acceptable: Lessons from a Wind Farm Project in
795 St-Valentin, Quebec. *Case Studies in the Environment*, 1.
796 https://doi.org/10.1525/cse.2018.001354

797 Steffen, W., Broadgate, W., Deutsch, L., Gaffney, O., & Ludwig, C. (2015). The
798 trajectory of the Anthropocene: The Great Acceleration. *The Anthropocene*
799 *Review*, *2*(1), 81-98. https://doi.org/10.1177/2053019614564785

800 Stevens, K., Harrisson, K. A., Hogan, F. E., Cooke, R., & Clarke, R. H. (2018).
801 Reduced gene flow in a vulnerable species reflects two centuries of habitat loss
802 and fragmentation. *Ecosphere*, *9*(2), e02114. https://doi.org/10.1002/ecs2.2114

803 Stewart, P. S., Voskamp, A., Santini, L., Biber, M. F., Devenish, A. J. M., Hof, C.,
804 Willis, S. G., & Tobias, J. A. (2022). Global impacts of climate change on avian
805 functional diversity. *Ecology Letters*, *25*(3), 673-685.
806 https://doi.org/10.1111/ele.13830

807 Thacker, S., Adshead, D., Fay, M., Hallegatte, S., Harvey, M., Meller, H.,
808 O'Regan, N., Rozenberg, J., Watkins, G., & Hall, J. W. (2019). Infrastructure for

809 sustainable development. *Nature Sustainability*, *2*(4), 324-331.
810 https://doi.org/10.1038/s41893-019-0256-8

811 Thaker, M., Zambre, A., & Bhosale, H. (2018). Wind farms have cascading
812 impacts on ecosystems across trophic levels. *Nature Ecology & Evolution*, *2*(12),
813 1854-1858. https://doi.org/10.1038/s41559-018-0707-z

814 Tong, D., Zhang, Q., Zheng, Y., Caldeira, K., Shearer, C., Hong, C., Qin, Y., &
815 Davis, S. J. (2019). Committed emissions from existing energy infrastructure
816 jeopardize 1.5 °C climate target. *Nature*, *572*(7769), 373-377.
817 https://doi.org/10.1038/s41586-019-1364-3

818 Townsend, J. M., McFarland, K. P., Rimmer, C. C., Ellison, W. G., & Goetz, J. E.
819 (2020). Bicknell's Thrush (*Catharus bicknelli*), version 1.0. *Birds of the World*.
820 https://doi.org/10.2173/bow.bicthr.01

821 Tremblay, J. A., Lessard, F., Riopel, M., Aubry, Y., & Desrochers, A. (2025).
822 Assessing *Catharus bicknelli* (Bicknell's Thrush) habitat dynamics: A high-
823 resolution model based on LiDAR metrics. *Ornithological Applications*, duaf040.
824 https://doi.org/10.1093/ornithapp/duaf040

825 van der Ree, R., Smith, D. J., & Grilo, C. (2015). The Ecological Effects of Linear
826 Infrastructure and Traffic. In *Handbook of Road Ecology* (p. 1-9). John Wiley &
827 Sons, Ltd. https://doi.org/10.1002/9781118568170.ch1

828 van Vliet, J. (2019). Direct and indirect loss of natural area from urban expansion.
829 *Nature Sustainability*, *2*(8), 755-763. https://doi.org/10.1038/s41893-019-0340-0

830 Venier, L. A., Thompson, I. D., Fleming, R., Malcolm, J., Aubin, I., Trofymow, J.
831 A., Langor, D., Sturrock, R., Patry, C., Outerbridge, R. O., Holmes, S. B.,
832 Haeussler, S., De Grandpré, L., Chen, H. Y. H., Bayne, E., Arsenault, A., &
833 Brandt, J. P. (2014). Effects of natural resource development on the terrestrial
834 biodiversity of Canadian boreal forests. *Environmental Reviews*, *22*(4), 457-490.
835 https://doi.org/10.1139/er-2013-0075

836 Venter, O., Sanderson, E. W., Magrach, A., Allan, J. R., Beher, J., Jones, K. R.,
837 Possingham, H. P., Laurance, W. F., Wood, P., Fekete, B. M., Levy, M. A., &
838 Watson, J. E. M. (2016). Sixteen years of change in the global terrestrial human
839 footprint and implications for biodiversity conservation. *Nature Communications*,
840 *7*(1), 12558. https://doi.org/10.1038/ncomms12558

841 Wimhurst, J. J., Nsude, C. C., & Greene, J. S. (2023). Standardizing the factors
842 used in wind farm site suitability models: A review. *Heliyon*, *9*(5).
843 https://doi.org/10.1016/j.heliyon.2023.e15903

844 Zurell, D., König, C., Malchow, A.-K., Kapitza, S., Bocedi, G., Travis, J., &
845 Fandos, G. (2022). Spatially explicit models for decision-making in animal
846 conservation and restoration. *Ecography*, *2022*(4).
847 https://doi.org/10.1111/ecog.05787

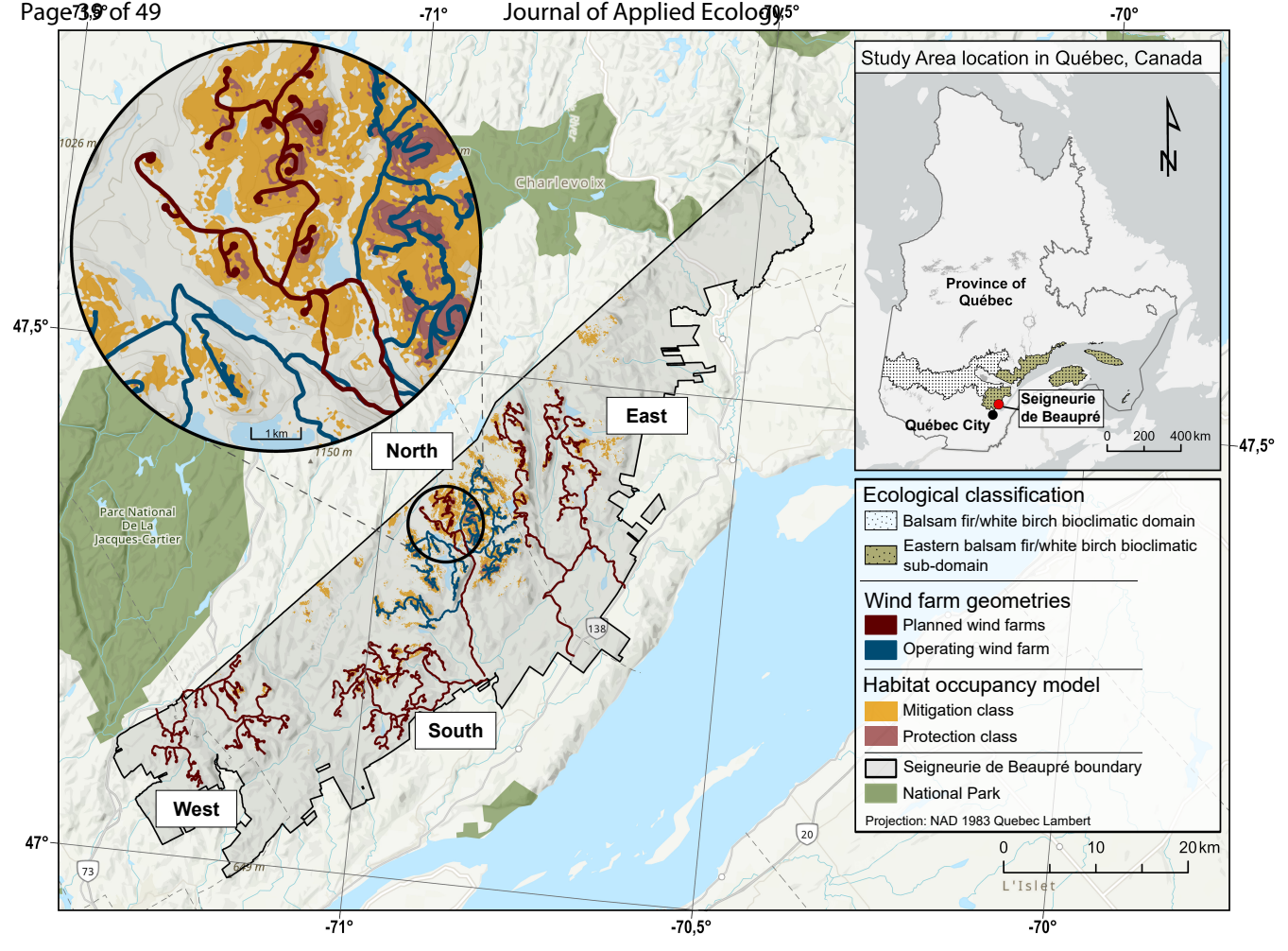


Figure 1. Study area and wind farm locations in the Seigneurie de Beaupré (Massif du Lac Jacques-Cartier, Québec, Canada), showing operating and planned wind farms and habitat classification based on the mitigation and protection thresholds from Tremblay *et al.* (2025).

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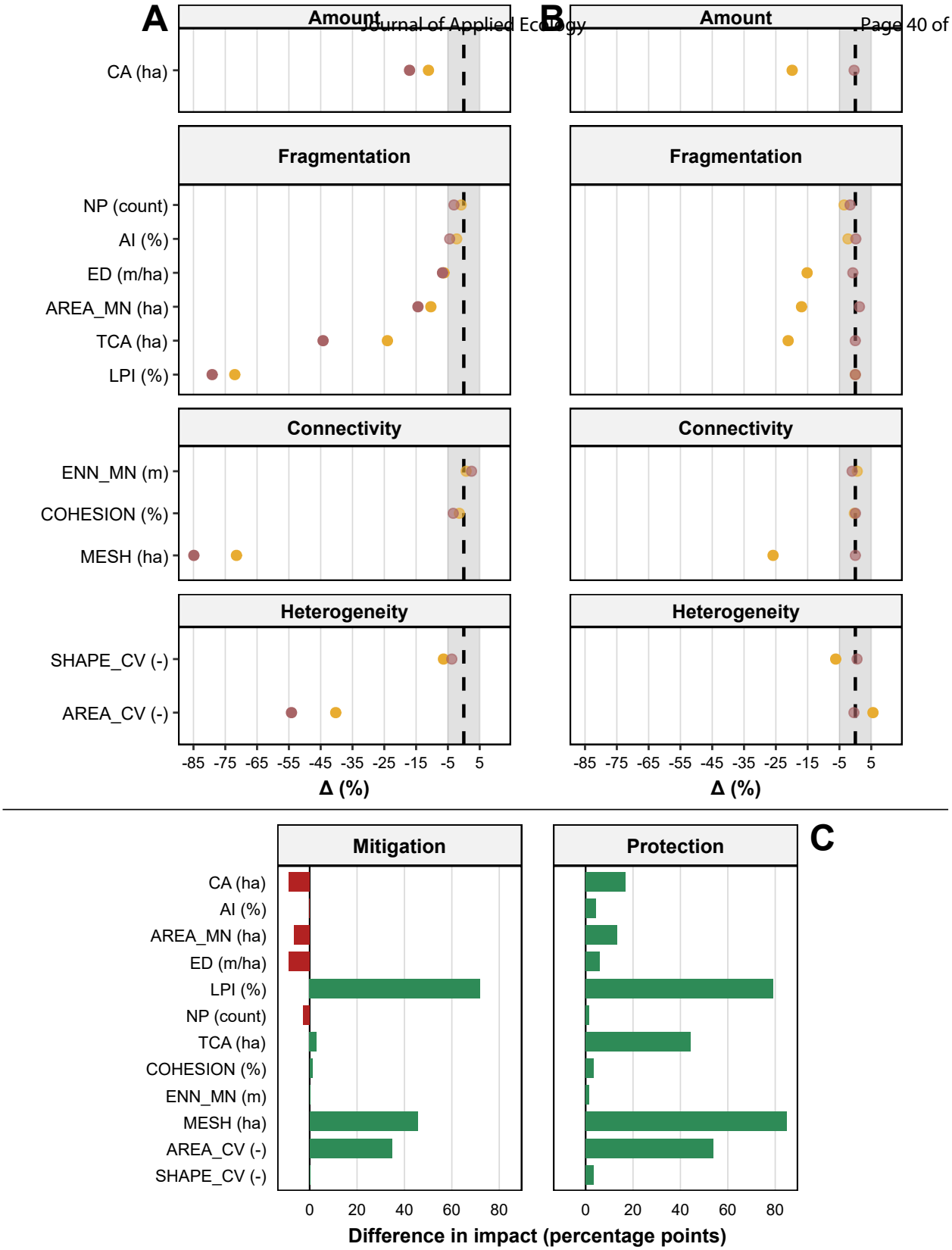
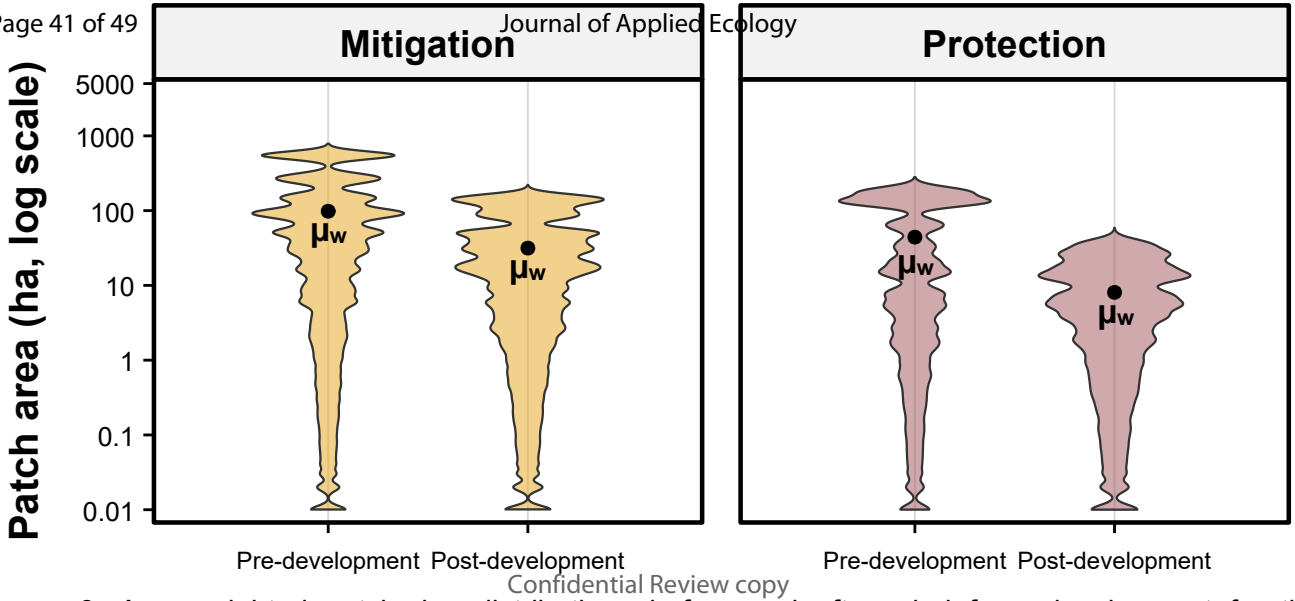


Figure 2. Relative changes in habitat amount and configuration metrics following **A)** wind farm development in the Seigneurie de Beaupré study area (Québec, Canada), and **B)** under the reference scenario (giving equal weight to ecological stake and wind opportunity), across the mitigation (yellow) and protection (brown) habitat classes. The direction and magnitude of change in metrics between **A)** and **B)** across both habitat classes, are shown in panel **C)** and indicated as increased (red) or reduced (green). For all panels, refer to Table 1 for the metric abbreviations.

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Figure 3. Area-weighted patch-size distributions before and after wind farm development for the mitigation and the protection habitat classes.

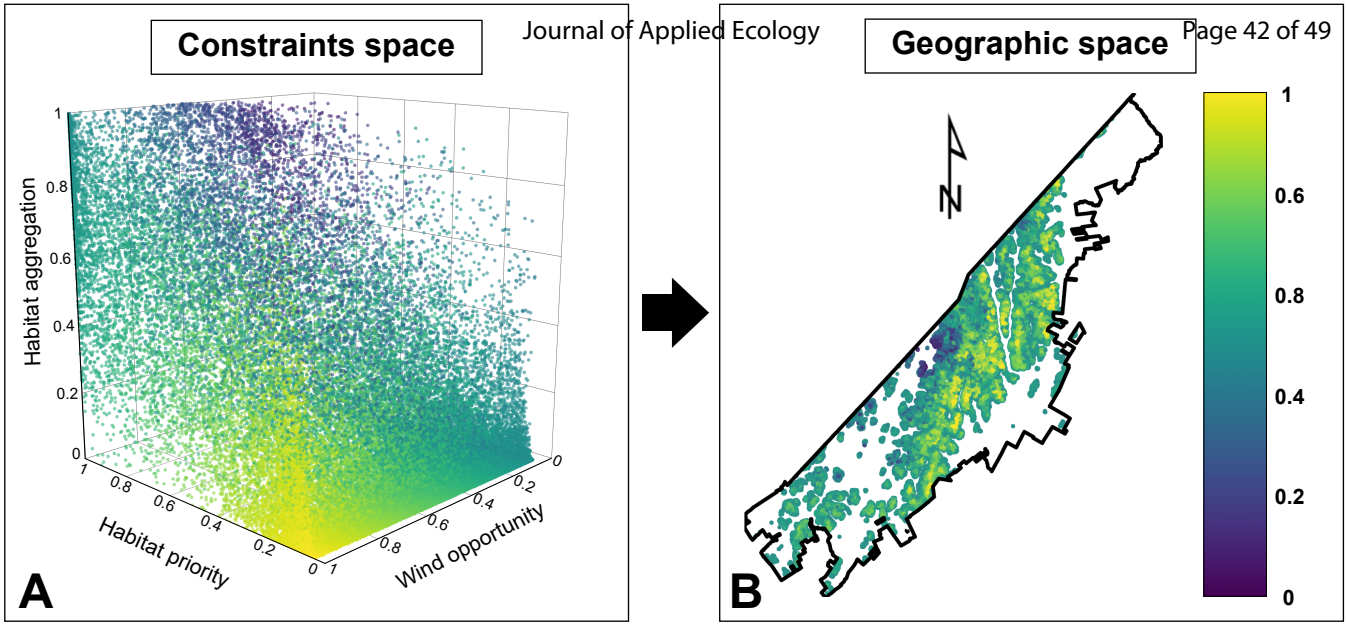


Figure 4. A) Three-dimensional constraints space showing the distribution of wind-eligible pixels according to habitat priority, wind opportunity, and local habitat aggregation under the *Reference* scenario, and **B)** their projection in geographic space within the study area.

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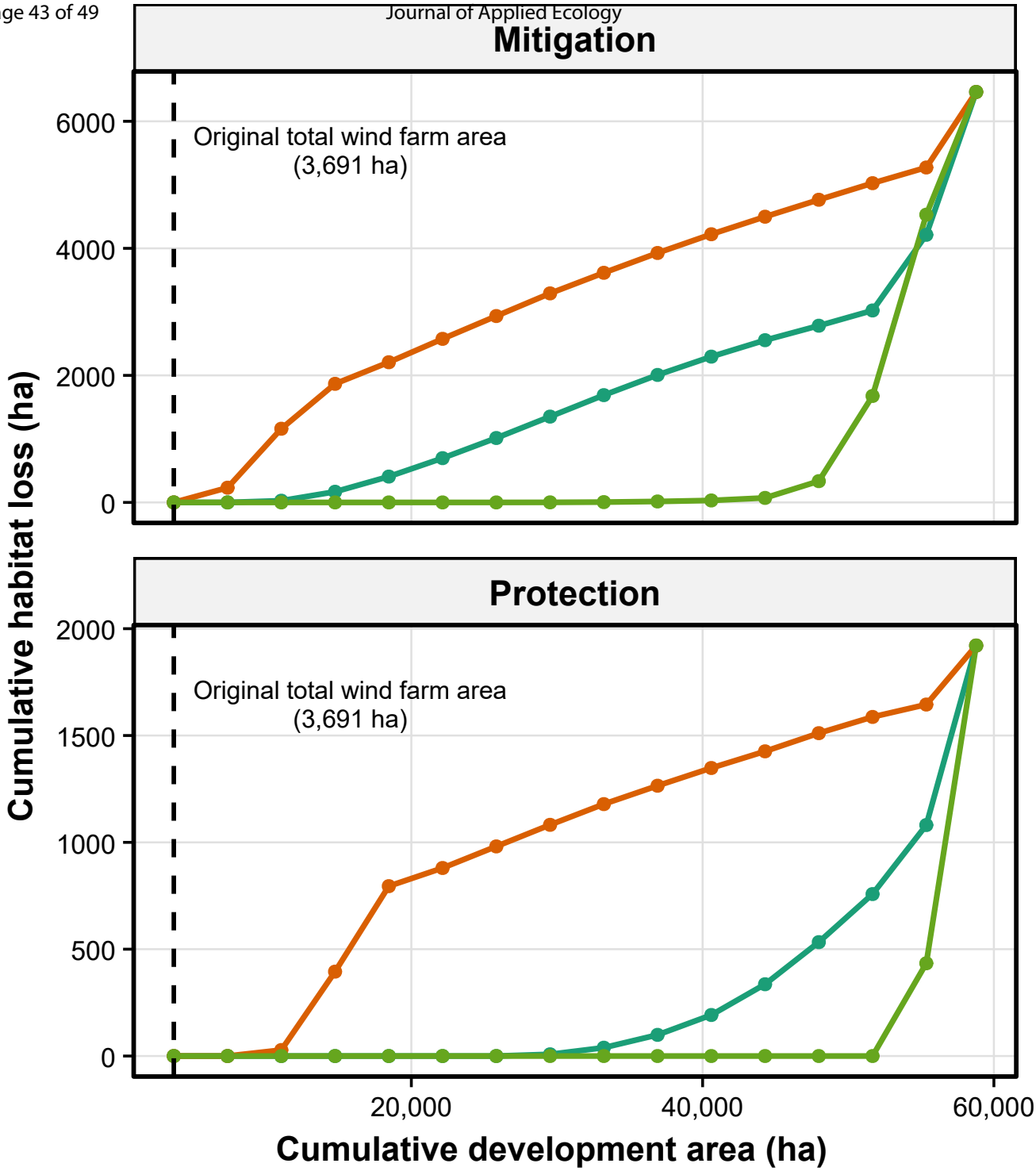


Figure 5. Cumulative habitat loss across scenarios for the mitigation and protection habitat classes. Each point represents one iteration, corresponding to the allocation of an additional area equivalent to the total original area of the four wind farms considered in this study (3,691 ha). Scenarios differ by the weighting applied to the three dimensions of the decision model (see Table 1): *Reference* (teal) represents a balanced trade-off; *Development* (orange) prioritizes wind opportunity; and *Conservation* (green) prioritizes the Bicknell's Thrush habitat constraints.

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Table 1. Landscape metrics used to quantify changes in habitat amount and configuration, including fragmentation, connectivity, and habitat heterogeneity.

Category	Metric	Abbreviation	Unit	Brief description
Habitat amount	Total class area	CA	ha	Total area occupied by the habitat class
Fragmentation	Number of patches	NP	count	Total number of patches
	Mean patch area	AREA_MN	ha	Average patch size
	Largest patch index	LPI	%	Percentage of the landscape occupied by the largest patch
	Edge density	ED	m/ha	Total edge length per unit area
	Aggregation index	AI	%	Degree of spatial aggregation among patches
	Total core area (30m)	TCA	ha	Core area excluding a 30 m edge buffer
Connectivity	Patch cohesion	COHESION	%	Physical connectedness of habitat patches
	Effective mesh size	MESH	ha	Surface-weighted mean patch size
	Mean Euclidean nearest-neighbor distance	ENN_MN	m	Mean distance to the nearest neighboring patch
Habitat heterogeneity	Coefficient of variation in patch area	AREA_CV	–	Relative variability in patch area
	Patch shape complexity	SHAPE_CV	–	Relative variability in patch shape complexity

Table 2. Weighting schemes used to combine the three dimensions of the decision model under alternative planning scenarios.

Scenario	Wind opportunity	Habitat priority	Habitat aggregation
Reference	0.50	0.35	0.15
Development-oriented	0.70	0.20	0.10
Conservation-oriented	0.30	0.50	0.20

Supplementary Material

Table S1. Wind farm development phases and their spatial parameters in the Seigneurie de Beaupré area.

Wind farm	Year of operation	Capacity (MW)	Number of turbines	Access roads corridor width (m)	Turbine pad area (ha)
North	2013-2015	364	164	40	0.5
South	Planned 2027	400	57	40	1.2
East	Planned 2028	400	57	40	1.2
West	Planned 2028	400	67	40	1.2

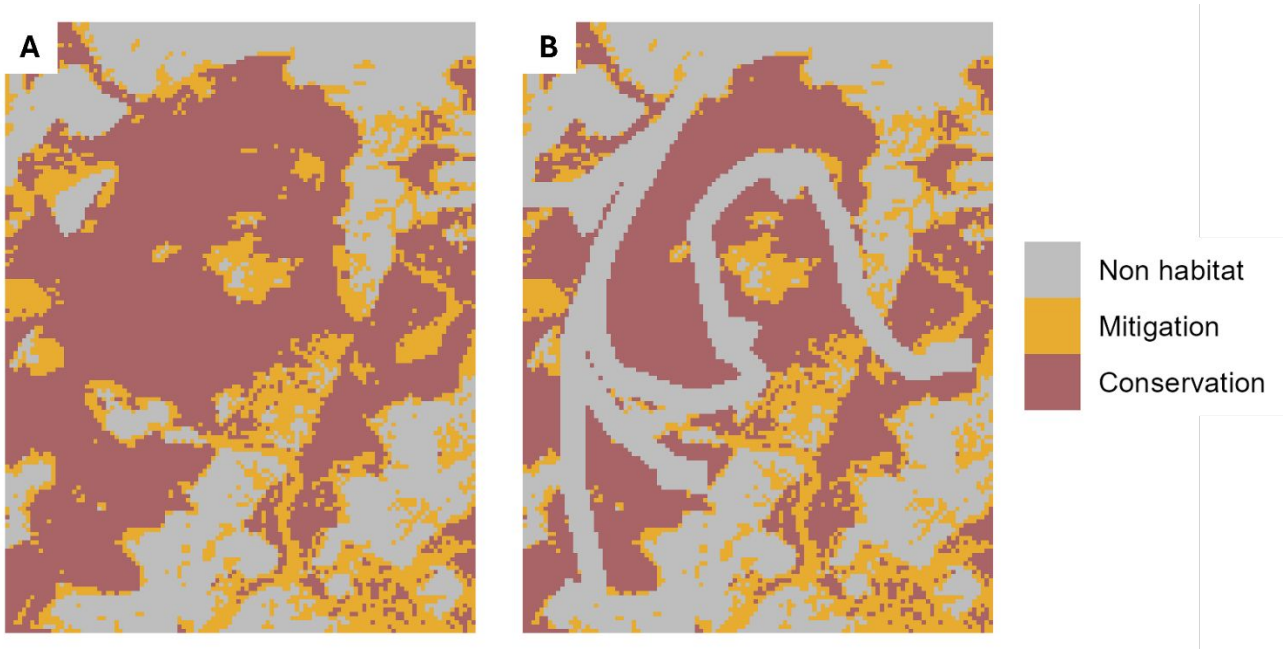


Figure S1. Comparison of **(A)** pre-development and **(B)** post-development habitat classes within the North wind farm sector, based on mitigation and protection thresholds.

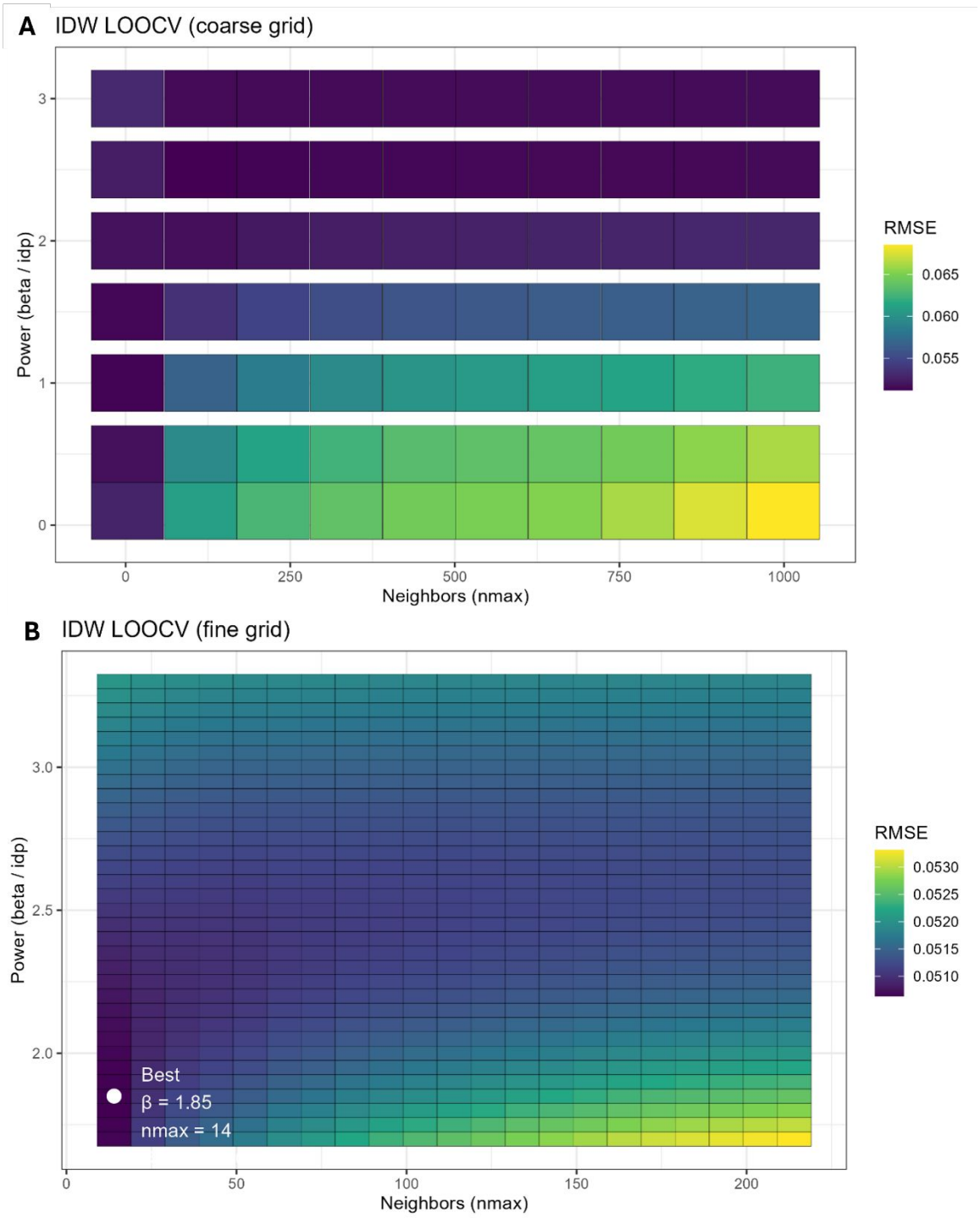


Figure S2. Leave-one-out cross-validation (LOOCV) optimization of inverse distance weighting (IDW) parameters. **(A)** Coarse and **(B)** fine grids show the RMSE surface across combinations of power (β, idp) and number of neighbors (n_{max}), with the optimal parameter set identified at the minimum RMSE.

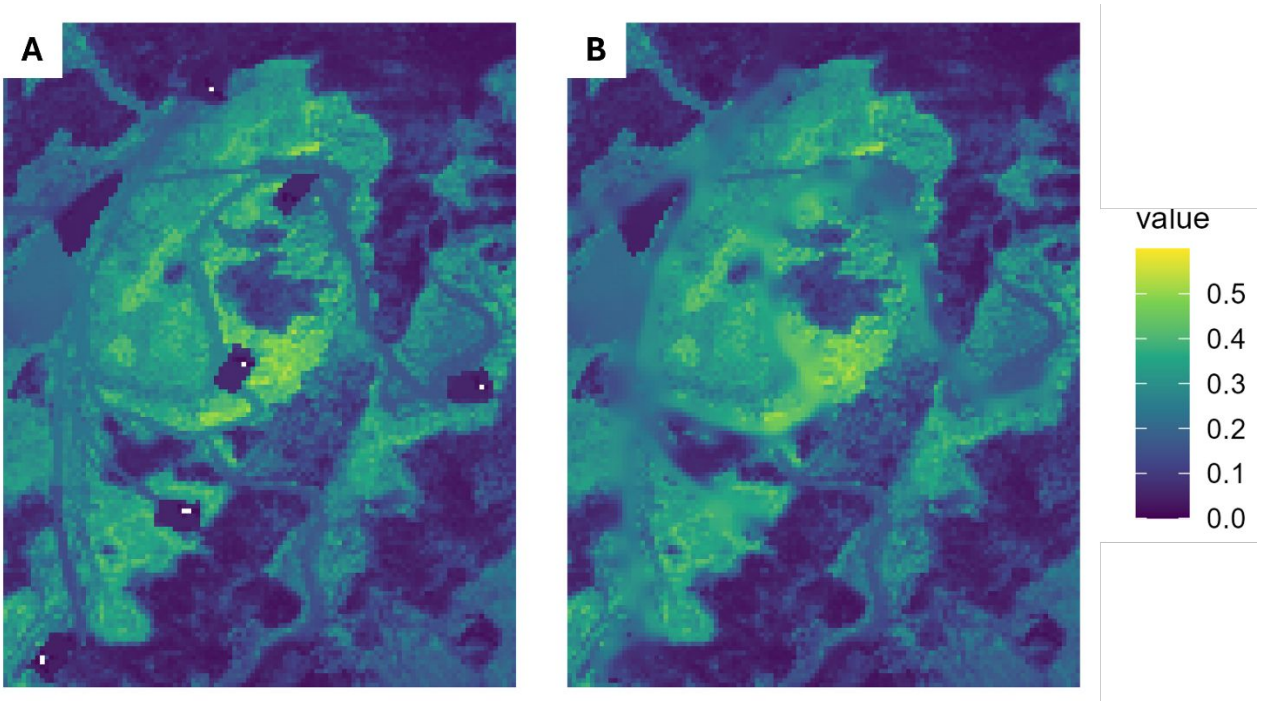


Figure S3. Comparison of model predictions **(A)** before and **(B)** after IDW reinterpolation within the North wind farm sector.